DO NOW — RECALL

Programming constructs and scope

Name: _____ Class: _____ Date: _____

1. What is the approximate growth of a loop executed n times?

2. Why must binary search update low=mid+1 rather than low=mid in the greater-than case?

3. What is true AND false?

4. What does int a=2; a=a+3; Console.WriteLine(a); output?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

O(n).

2. Model answer

It must exclude mid and shrink the search interval.

3. Model answer

false.

4. Model answer

5.

Vocabulary — know these terms

sequence — Operations execute in order.

selection — Execution chooses a branch according to a condition.

iteration — A block repeats while a condition permits.

scope — Where a name can be accessed.

shadowing — A local declaration hides another accessible name.

NEW LEARNING

Read, annotate and explain

Three control structures

Sequence runs in order; selection chooses a branch; iteration repeats instructions.

Where is a variable visible?

A local declaration belongs to its block or method; a class-level field can be accessed by relevant class methods.

Shadowing

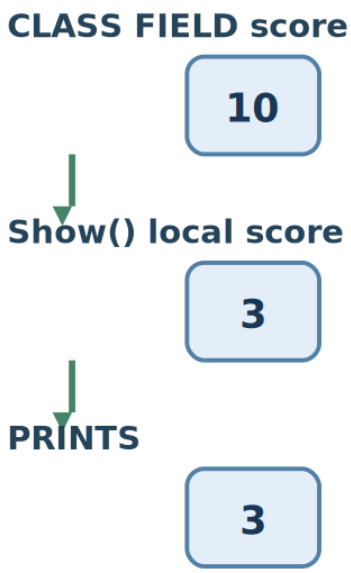
A local variable with the same name as a field can hide that field inside the method; trace precisely which one is printed.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read the code; predict its effect



Worked algorithm states: CLASS FIELD score 10; Show() local score 3; PRINTS 3

I DO — WORKED EXAMPLE

Follow the worked method

A class field x is 8. A method declares int x=2 and prints x. What appears?

Predict first

Which x is in scope at the print statement?

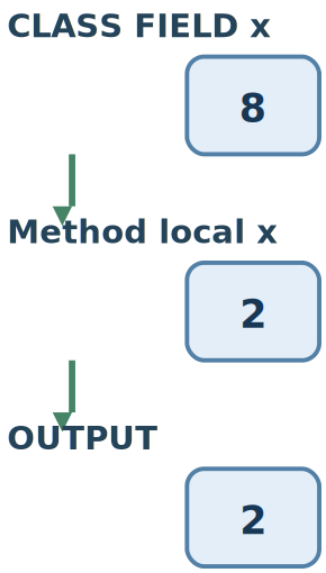
Model solution

The method-local x is 2 and shadows the field. The program prints 2.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A class field x is 8. A method declares int x=2 and prints x. What appears?



Worked algorithm states: CLASS FIELD x 8; Method local x 2; OUTPUT 2

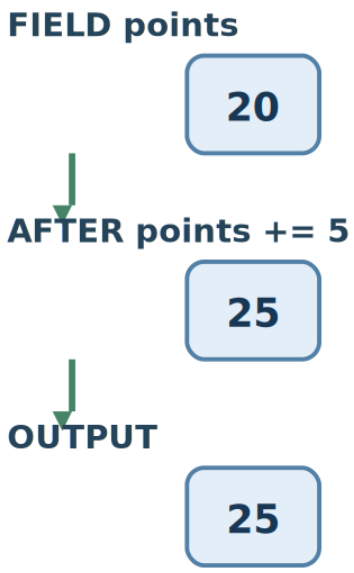
WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A class field points is 20. A method runs points += 5 and prints points without declaring a local variable.



Worked algorithm states: FIELD points 20; AFTER points += 5 25; OUTPUT 25

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Which variable is modified?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Class field level = 9; method declares int level = 4 and prints level. What appears?

Question 2

Class field attempts = 2; method executes attempts++; then prints attempts. What appears?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Describe one risk of unnecessary global/class-level mutable variables.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A local variable always changes a same-named class field.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model solution

The class field is modified to 25, then 25 is printed.

IMPROVE: Explain one correction to your own reasoning.